1 Introduction to R and RStudio

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1 Welcome to R Programming!

1.1 What is R?

R is a powerful programming language and environment for:

- Statistical computing and analysis
- Data visualization
- Data manipulation and cleaning
- Machine learning and predictive modeling
- Reproducible research

1.2 Why Learn R?

- **Free and Open Source**: No licensing costs
- **Extensive Packages**: Over 19,000 packages available
- **Strong Community**: Active support and resources
- **Industry Standard**: Used in academia, healthcare, finance, tech

- **Data Science Ready**: Built specifically for data analysis

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2 Getting Started with RStudio

2.1 RStudio Interface Overview

RStudio has 4 main panes:

1. **Source/Editor Pane** (Top Left): Where you write and save your code
2. **Console Pane** (Bottom Left): Where code executes and shows output
3. **Environment/History Pane** (Top Right): Shows variables and command history
4. **Files/Plots/Packages/Help Pane** (Bottom Right): Multiple tabs for different functions

2.1.1 Quick Tips:

- Press **Ctrl + Enter** (Windows) or **Cmd + Enter** (Mac) to run current line
- Press **Ctrl + Shift + Enter** to run entire script
- Use # for comments (code that won't execute)

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3 Your First R Code

3.1 The Classic "Hello World"

```
print("Hello World")
```

```
[1] "Hello World"
```

3.2 Built-in Constants

```r
# Pi is a built-in constant in R
pi
```

```
[1] 3.141593
```

```r
# R has many mathematical constants
exp(1)  # Euler's number (e)
```

```
[1] 2.718282
```

---

4 Basic Math Operations

4.1 Arithmetic Operators

```r
# Addition
2 + 2
```

```
[1] 4
```

```r
# Subtraction
10 - 3
```

```
[1] 7
```

```r
# Multiplication
4 * 5
```

```
[1] 20
```

```r
# Division
20 / 4
```

```
[1] 5
```

```r
# Exponentiation (Power)
2^3
```

```
[1] 8
```

```r
# Modulo (Remainder)
10 %% 3
```

```
[1] 1
```

```r
# Integer Division
10 %/% 3
```

```
[1] 3
```

4.2 Order of Operations (PEMDAS)

```r
# R follows standard mathematical order
2 + 3 * 4      # Multiplication first
```

```
[1] 14
```

```r
# Use parentheses to change order
(2 + 3) * 4    # Addition first
```

```
[1] 20
```

---

5 Data Types in R

5.1 1. Numeric (Double)

The default for numbers with decimals:

```r
x <- 5
x
```

```
[1] 5
```

```r
class(x)
```

```
[1] "numeric"
```

```r
typeof(x)
```

```
[1] "double"
```

```r
# Decimal numbers
m <- 0.03
class(m)
```

```
[1] "numeric"
```

```r
# Scientific notation
large_num <- 1.5e6  # 1,500,000
large_num
```

```
[1] 1500000
```

5.2 2. Integer

Whole numbers (add 'L' suffix):

```r
y <- 444L
class(y)
```

```
[1] "integer"
```

```r
typeof(y)
```

```
[1] "integer"
```

```r
# Check if it's an integer
is.integer(y)
```

```
[1] TRUE
```

```r
is.integer(5)      # FALSE - default is numeric
```

```
[1] FALSE
```

```r
is.integer(5L)     # TRUE
```

```
[1] TRUE
```

5.3 3. Character (String)

Text data enclosed in quotes:

```r
z <- "This is our first data camp"
z
```

```
[1] "This is our first data camp"
```

```r
class(z)
```

```
[1] "character"
```

```r
# Single or double quotes both work
name <- 'John'
greeting <- "Hello, World!"

# Combine strings
paste("Hello", "R", "Programming")
```

```
[1] "Hello R Programming"
```

```r
paste0("Hello", "World")  # No space
```

```
[1] "HelloWorld"
```

5.4 4. Logical (Boolean)

TRUE or FALSE values:

```r
aa <- FALSE
class(aa)
```

```
[1] "logical"
```

```r
bb <- TRUE
bb
```

```
[1] TRUE
```

```r
# Logical operations
TRUE & FALSE   # AND
```

```
[1] FALSE
```

```r
TRUE | FALSE   # OR
```

```
[1] TRUE
```

```r
!TRUE          # NOT
```

```
[1] FALSE
```

```r
# Comparisons return logical values
5 > 3
```

```
[1] TRUE
```

```r
10 == 10
```

```
[1] TRUE
```

```r
7 != 8
```

```
[1] TRUE
```

5.5 Quick Type Check Summary

```r
# Check multiple types at once
check_value <- 42

is.numeric(check_value)
```

```
[1] TRUE
```

```r
is.character(check_value)
```

```
[1] FALSE
```

```r
is.logical(check_value)
```

```
[1] FALSE
```

---

6 Variables and Assignment

6.1 Creating Variables

```r
# Use <- or = for assignment (← is preferred in R)
k <- 11
l <- 9

# View variables
k
```

```
[1] 11
```

```
l
```

```
[1] 9
```

```
# Use variables in calculations
k + l
```

```
[1] 20
```

```
k * l
```

```
[1] 99
```

```
k / l
```

```
[1] 1.222222
```

```
# Update variables
k <- k + 5
k
```

```
[1] 16
```

6.2 Variable Naming Rules

```
# Good variable names
student_age <- 20
studentAge <- 20      # camelCase
student.age <- 20     # dot notation

# Names can include numbers (but not start with them)
data1 <- 100
# 1data <- 100  # This would cause an ERROR

# Case sensitive
Value <- 10
value <- 20
Value == value  # FALSE
```

```
[1] FALSE
```

6.3 Best Practices

- Use descriptive names: `student_score` instead of `x`
- Be consistent with naming style
- Avoid special characters except `.` and `_`
- Don't overwrite built-in functions (like `mean`, `sum`, `c`)

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7 Using Functions

7.1 What are Functions?

Functions are pre-built commands that perform specific tasks.

Syntax: `function_name(arguments)`

7.2 Basic Mathematical Functions

```
# Square root
sqrt(16)
```

```
[1] 4
```

```
# Absolute value
abs(-5)
```

```
[1] 5
```

```
# Rounding
round(3.14159, 2)  # Round to 2 decimal places
```

```
[1] 3.14
```

```
# Logarithms
log(10)      # Natural log
```

```
[1] 2.302585
```

```
log10(100)   # Base 10 log
```

```
[1] 2
```

```
# Exponential
exp(2)
```

```
[1] 7.389056
```

7.3 The c() Function (Concatenate)

Creates vectors (collections of values):

```
# Create a vector of numbers
numbers <- c(4, 5, 6)
numbers
```

```
[1] 4 5 6
```

```
# Calculate mean
mean(c(4, 5, 6))
```

```
[1] 5
```

```
# Calculate sum
sum(c(4, 5, 6, 7))
```

```
[1] 22
```

```
# Create sequences
sum(1:10)      # Sum of 1 to 10
```

```
[1] 55
```

```r
mean(1:10)     # Mean of 1 to 10
```

```
[1] 5.5
```

```r
# More with vectors
ages <- c(23, 45, 67, 34, 29)
min(ages)
```

```
[1] 23
```

```r
max(ages)
```

```
[1] 67
```

```r
median(ages)
```

```
[1] 34
```

```r
length(ages)   # How many elements
```

```
[1] 5
```

7.4 Statistical Functions

```r
# Create sample data
scores <- c(85, 90, 78, 92, 88, 76, 95)

# Descriptive statistics
mean(scores)       # Average
```

```
[1] 86.28571
```

```r
median(scores)     # Middle value
```

```
[1] 88
```

```r
sd(scores)          # Standard deviation
```

```
[1] 7.087884
```

```r
var(scores)         # Variance
```

```
[1] 50.2381
```

```r
range(scores)       # Min and max
```

```
[1] 76 95
```

```r
summary(scores)     # Quick summary
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  76.00   81.50   88.00   86.29   91.00   95.00 
```

---

8 Sequences and Patterns

8.1 Creating Sequences

```r
# Using colon operator
1:10
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```r
5:15
```

```
 [1]  5  6  7  8  9 10 11 12 13 14 15
```

```r
10:1  # Descending
```

```
 [1] 10  9  8  7  6  5  4  3  2  1
```

```r
# Using seq() function
seq(1, 10)                    # Same as 1:10
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```r
seq(0, 100, by = 10)          # Specify step
```

```
 [1]   0  10  20  30  40  50  60  70  80  90 100
```

```r
seq(0, 1, by = 0.1)           # Decimal steps
```

```
 [1] 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
```

```r
seq(0, 1, length.out = 11)    # Specify length
```

```
 [1] 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
```

```r
# Repeating values
rep(5, times = 3)             # Repeat 5 three times
```

```
[1] 5 5 5
```

```r
rep(1:3, times = 3)           # Repeat sequence
```

```
[1] 1 2 3 1 2 3 1 2 3
```

```r
rep(1:3, each = 3)            # Repeat each element
```

```
[1] 1 1 1 2 2 2 3 3 3
```

---

9 Getting Help in R

9.1 Help Functions

```r
# Get help on a function
?mean
help(sum)

# Search for topics
??regression
help.search("linear model")

# Examples of function usage
example(mean)

# See arguments of a function
args(seq)
```

9.2 Useful Commands

```r
# List objects in environment
ls()
```

```
 [1] "aa"              "ages"            "bb"              "check_value"
 [5] "data1"           "greeting"        "k"               "l"
 [9] "large_num"       "m"               "name"            "numbers"
[13] "pandoc_dir"      "quarto_bin_path" "scores"          "student_age"
[17] "student.age"     "studentAge"      "value"           "Value"
[21] "x"               "y"               "z"
```

```r
# Remove an object
# rm(x)

# Clear entire environment
# rm(list = ls())

# Check current working directory
getwd()
```

```
[1] "/home/mk/Documents/khanalmohan.github.io/r-tutorial"
```

```
# Set working directory (example - modify path)
# setwd("C:/Users/YourName/Documents/R_Training")
```

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10 Practice Exercises

10.1 Exercise 1: Basic Calculations

Calculate the following and store in variables:

1. The sum of 15 and 27
2. 8 raised to the power of 3
3. The square root of 144

```
# Your code here
sum_result <- 15 + 27
power_result <- 8^3
sqrt_result <- sqrt(144)

# View results
sum_result
```

```
[1] 42
```

```
power_result
```

```
[1] 512
```

```
sqrt_result
```

```
[1] 12
```

10.2 Exercise 2: Working with Vectors

Create a vector of temperatures: 72, 68, 75, 80, 77

Calculate: - Mean temperature - Highest temperature - Temperature range

```r
# Your code here
temperatures <- c(72, 68, 75, 80, 77)

mean(temperatures)
```

```
[1] 74.4
```

```r
max(temperatures)
```

```
[1] 80
```

```r
range(temperatures)
```

```
[1] 68 80
```

10.3 Exercise 3: Data Types

Identify the data type of the following:

```r
var1 <- 42
var2 <- "Data Science"
var3 <- TRUE
var4 <- 3.14159

class(var1)
```

```
[1] "numeric"
```

```r
class(var2)
```

```
[1] "character"
```

```r
class(var3)
```

```
[1] "logical"
```

```r
class(var4)
```

```
[1] "numeric"
```

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11 Key Takeaways

1. **R is a powerful tool** for data analysis and statistics
2. **RStudio** provides an excellent interface for R programming
3. **Data types** matter: numeric, integer, character, logical
4. **Variables** store values using `<-` operator
5. **Functions** perform operations: `mean()`, `sum()`, `c()`, etc.
6. **Vectors** are collections of values created with `c()`
7. **Help is available** using `?function_name`

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12 Homework / Practice

Before our next session, practice:

1. Install R and RStudio on your computer (if not already done)
2. Create 5 different variables with different data types
3. Perform at least 10 different mathematical operations
4. Create 3 vectors and calculate their mean, median, and standard deviation
5. Experiment with the `seq()` and `rep()` functions
6. Use the `?` operator to explore 3 new functions

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13 Next Session Preview

January 07, 2026: Data Import and Data Wrangling

We'll cover: - Reading CSV, Excel, and other data files - Introduction to the `dplyr` package - Data cleaning and manipulation techniques - Handling missing data (NA values)

Prepare: Make sure you have a CSV file ready to import (any data will work!)

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14 Resources

- R Documentation
- RStudio Cheatsheets
- R for Data Science (Free Book)
- Stack Overflow - R Tag

Questions? Feel free to ask during the session or practice at home!

Happy Coding!